MILITARY SPECIFICATION

DETONATOR, ELECTRIC, PA526 (XM70) LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, and packaging for Detonator, Electric, PA526 (XM70) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government drawings, documents, or publications. The following Government drawings, documents, or publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9311162 - Waterproof Sealant
9362755 - Detonator, Electric, PA526 (XM70)
9362756 - Bridge/Plug Assembly
9362757 - Plug Assembly
9362758 - Cup, Charged
9362759 - Insulator
9362760 - Sleeve
9362761 - Cup
9362762 - Electrode

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM Method E-300 - Recommended Practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Transportation

(The code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.) Orders should specify "CFR 49, Parts 100-199 (latest revision).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications contained therein and in this detailed specification.

3.2 Loading. The loading of the detonator shall comply with all requirements specified in the applicable drawings and specifications.

3.3 Detailed requirements.

3.3.1 Moisture content of lead azide. The moisture content of the lead azide at the loading station at the time of loading the detonators, shall not exceed 0.50 percent when determined as specified in 4.5.1.

3.3.2 Moisture content of RDX. The moisture content of the RDX, at the loading station at the time of loading the detonators, shall not exceed 0.50 percent when determined as specified in 4.5.1.

3.3.3 Resistances of bridge/plug assembly. Resistances of the bridge assembly shall comply with the requirements of the applicable drawing (see Dwg. 9362756).

3.3.4 Resistances of the detonator. Resistances of the detonator shall comply with requirements of the applicable drawing (see Dwg. 9362755).

3.3.5 Waterproofness. The detonator shall be able to function satisfactorily to the requirements given in 3.3.6 after being subjected to water immersion as specified in 4.5.5.

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3.3.6 Functioning.

3.3.6.1 Time/duds. The detonator shall fire on the first attempt, within eight (8) microseconds when tested as specified in 4.5.4.1.

3.3.6.2 Output. The detonator shall cause a depression in the test block of not less than 0.254mm (0.010 inches) when tested as specified in 4.5.4.2.

3.3.6.3 Non-functioning. The detonator shall not fire when tested in accordance with test conditions specified in 4.5.7.

3.3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.3.8 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product are not compromised by foreign material or any condition which may degrade the item. The following also applies:

All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Particular attention shall be given to adequacy and neatness of the soldering operations. There shall be no evidence of burning or charring.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for valuation in accordance with provisions of 4.3.2. The first article sample shall consist of twenty five (25) complete sets of parts and sub-assemblies and 185 complete detonator assemblies.

The samples shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same source as will be used in regular production.

4.3.2 Inspections to be performed. See MIL-A-48078. The sample shall be subjected by the Government to any or all of the examinations or tests specified in this specification (see Table 1).

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70432

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Detonator, Electric, PA526, Inert Parts and Complete Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg 9362761)				
	Examination	25			4.4.2.1
	Sleeve (Dwg 9362760)				
	Examination	25			4.4.2.2
	Plug assembly (Dwg 9362757)				
	Examination	25			4.4.2.3
	Bridge/plug assembly (Dwg 9362756)				
	Examination	25			4.4.2.4
	Resistance Test	50(a)			4.4.3.2
	Detonator assembly (Dwg 9362755)				
	Examination	50(b)			4.4.2.5
	Resistance Test	50(b)			4.4.3.3
	Functioning Time Test	35			4.4.3.4.1
	Waterproofness Test	50			4.4.3.4.2
	Dud Test	100			4.4.3.4.3

NOTES:
(a) The 25 samples for examination may be combined with an additional 25 samples for the total of 50 samples.
(b) Randomly selected from the 185 samples to be functionally tested.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of the detonator shall contain:

a. Metal parts of one interfix lot number from one manufacturer.

b. RDX and lead azide of one interfix lot number from one manufacturer.

The lot shall comply with the provisions for submission of product as specified in MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105, Inspection Level II, as applicable.

b. Automatic gaging. Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capabilty in accordance with contract.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70432

PARAGRAPH	TITLE	SHEET 1 OF 10	DRAWING NUMBER
4.4.2.1	Cup		9362761
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Thickness through bottom, minimum		0.65%		Gage
102	Inside diameter		0.65%		Gage
103	Total length, minimum		0.65%		Gage
104	Cup cracked or dented		0.65%		Visual
Minor					
201	Outside diameter, minimum		1.5%		Gage
202	Interior finish improper		1.5%		Visual
203	Evidence of poor workmanship		0.5%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 10		DRAWING NUMBER
4.4.2.2	Sleeve				9362760
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter, minimum		0.40%		Gage
102	Length		0.40%		Gage
Minor					
201	Inside diameter		0.65%		Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70432

PARAGRAPH	TITLE		SHEET 3 OF 10		DRAWING NUMBER
4.4.2.3	Plug assembly				9362757
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Distance between leads at glass surface on both sides of sleeve		0.40%		Gage
102	Evidence of poor workmanship on bridge surface of glass between leads		0.40%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 10		DRAWING NUMBER 9362756
4.4.2.4	Bridge/plug assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Distance from end of each lead to sleeve		0.65%		Gage
102	Bridge wire missing or damaged		0.65%		Visual
103	Color code on lead missing or incorrect		0.65%		Visual
104	Coating or plating on sleeve or leads missing or inadequate (bare spots or scratches)		0.65%		Visual
Minor					
201	Evidence of poor workmanship		1.5%		Visual
202	Bridge wire extends beyond soldering mound		1.5%		Visual
203	Soldering defective		1.5%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 10		DRAWING NUMBER
4.4.2.5	Detonator assembly				9362755
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
001	Ends of wires not twisted or inadequately twisted		1.00%		Visual
Major					
101	Outside diameter of cup, maximum		1.00%		Gage
102	Length of cup		1.00%		Gage
103	Cup cracked, dented or distorted		1.00%		Gage
104	Cup-sleeve joint seal missing or inadequate		1.00%		Visual
105	Cup crimp over sleeve missing or less than 360 degrees		1.00%		Visual
106	Lead wire broken or with deep cut		1.00%		Visual
Minor					
201	Evidence of poor workmanship (see 3.3.8)		.65%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 6 OF 10		DRAWING NUMBER
4.4.2.6	Bundle (prior to insertion into carton)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Incorrect number of detonators		0.65%		Visual
102	Lead wires not wrapped around the bundle or inadequately securing the bundle		0.65%		Visual
103	Foil wrapping incorrect or damaged exposing detonators		0.65%		Visual
Minor	None Defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70432

PARAGRAPH	TITLE	SHEET 7 OF 10	DRAWING NUMBER
4.4.2.7	Pocket assembly (prior to folding lip)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Detonator missing		0.65%		Visual
102	Adhesive seal inadequate (foil separating)		0.65%		Visual
103	Detonator inverted		0.65%		Visual
Minor	None Defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 8 OF 10		DRAWING NUMBER
4.4.2.8	Pocket assembly (prior to inserting into carton)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Pocket assembly damaged exposing detonator		0.40%		Visual
102	Lip not folded or folding inadequate		0.40%		Visual
Minor	None Defined				

NOTES:

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PARAGRAPH	TITLE		SHEET 9 OF 10		DRAWING NUMBER
4.4.2.9	Carton, sealed (prior to insertion into barrier bag)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Box damaged exposing contents		0.40%		Visual
Minor					
201	Contents loose		1.50%		Manual
202	Marking misleading or unidentifiable, including incorrect color code and patent marking		1.50%		Visual
203	Label missing or badly wrinkled		0.50%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 10 OF 10		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
4.4.2.10	Barrier bag, sealed				
CATEGORY	**EXAMINATION OR TEST**	**NO. OF SAMPLE UNITS**	**AQL OR 100%**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE /INSPECTION METHOD**
Critical	None Defined				
Major 101	Waterproofness destroyed by separation of heat seal, or rip, tear or puncture		0.40%		Visual
Minor 201	Marking misleading or unidentifiable, including incorrect color code		0.40%		Visual

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4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures, including use of proper barricades, must be followed in handling and processing explosives and detonators.

4.4.3.1 Moisture content of lead azide and RDX. Obtain two independent, representative samples of each of the explosives using a procedure prescribed in ASTME-300 for sampling solid materials. This determination shall be made at the site of, and immediately prior to loading the cups. If any sample fails to comply with the requirements of paragraph 3.3.1 or 3.3.2, the material shall not be used until further dried to meet the requirements upon re-testing. If loading has begun, all detonators loaded with the questionable material shall be rejected.

4.4.3.2 Resistance of bridge assembly. These resistance measurements shall be performed 100 percent. Bridge assemblies failing to meet the specified requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.5.2.

4.4.3.3 Resistances of detonator (prior to twisting leads). These resistance measurements shall be performed 100 percent. Detonators failing to meet the specified requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.5.3.

4.4.3.4 Functioning and waterproofness. A sample of 185 detonators shall be selected from each lot and the sample apportioned as specified below for testing in the sequence specified below. The tests shall be conducted at a temperature of 21 $\pm$ 5°C (70 $\pm$ 10°F).

4.4.3.4.1 Duds. A sample of 35 detonators shall be subjected to the functioning time test specification in 4.5.4.1 using an approved test equipment. The lot shall be rejected if the mean functioning time, ($\bar{X}$) plus the product of 2.27 and the standard deviation (s) of the sample exceeds 8 microseconds. Detonators that fail to fire shall not be considered in the required calculation, but will be replaced with detonators from the remaining portion of the original sample. If three duds are encountered while performing this test, the lot shall be rejected and testing for acceptance by the Government shall be stopped. Subsequent testing for waterproofness, duds, and output shall not be conducted. The standard deviation shall be calculated using the formula in 6.5, or its equivalent.

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4.4.3.4.2 Waterproofness, output, and duds. If the lot has not been rejected in 4.4.3.4.1 a sample of 50 detonators shall be tested as specified in 4.5.5 using an approved test equipment (see 4.5.4). The lot shall be rejected if two or more detonators fail to produce a depression in the block of the required size. Detonators that fail to fire shall be replaced with detonators from the remaining portion of the original sample. If a total of three duds has been encountered during the time and output tests combined, the lot shall be rejected and further testing for acceptance by the Government shall be stopped. If a total of three duds has not been encountered but one detonator fails to create a depression in the block of the required depth, the balance of the original sample shall be tested. The lot shall be rejected and testing for Government acceptance shall be stopped if one or more detonators fail to create a depression in the block of the required depth, a total of three duds is encountered in all the combined testing.

4.4.3.4.3 Duds. If all of the first 50 detonators tested in 4.4.3.4.2 function satisfactorily in the output phase, and the lot has not been rejected for duds, the remaining portion of the original sample shall be subjected to the test for duds specified in 4.5.6 using an approved test equipment (see 4.5.4). The lot shall be rejected when and if a total of three duds is encountered in all of the combined testing, and testing for Government acceptance shall be stopped. If two duds are encountered, a second sample of 420 detonators shall be selected from the lot and subjected to the test specified in 4.5.6. Testing shall be stopped for Government acceptance and the lot shall be rejected if and when the total of duds in all combined testing reaches three.

4.4.3.5 Non-functioning. This test shall be performed 100 percent. Detonators failing to meet the specified requirement shall be classed defective. The test shall be performed as specified in 4.5.7 using an approved test equipment in accordance with 4.4.4.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures (see 6.4).

4.5.1 Test method for moisture content of lead azide and RDX. Moisture content of the explosives shall be determined in accordance with Method 101.6, MIL-STD-650.

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4.5.2 Resistance of bridge assembly. Using the specified resistance measuring equipment, the bridge assembly shall be tested to determine compliance with the two (2) resistance requirements specified on the applicable drawing (Dwg 9352756).

4.5.3 Resistances of detonator.

WARNING - This test shall be performed behind a suitable barricade.

Prior to twisting (shunting) the lead wires, the resistances of detonator shall be measured with an approved electrical ohmmeter. The current passing through the detonator shall not exceed one hundred (100) microamperes.

4.5.4 Functioning. The equipment used for the following tests requires prior approval of the Technical Agency.

4.5.4.1 Functioning time. The detonator shall be assembled to a test fixture and connected electrically to the test circuit. The energy from a capacitor of sixteen (16) microfarads, max, charged to 200 volts, max, shall be discharged to fire the detonator. Accurately measure the time interval between the application of energy and the functioning of the detonator. Record the time. Also, record any incidence of duds.

4.5.4.2 Output. This test may be performed concurrently with functioning time test given in 4.5.4.1. The test shall be performed using a steel disc with a Rockwell Hardness of B90 (Test 30.1.1, MIL-STD-331). Measure the depression made in the steel disc and record.

4.5.5 Waterproofness test method. Submerge the detonators in water at a depth of not less than 5.08cm (two iches) but not more than 7.62cm (three inches) for a period of 48 hours minimum. The water temperature shall be maintained at 20 $\pm$ 2°C (68°F). The detonators shall be tested to determine conformance with 3.3.6 within 30 minutes after removal from the water.

4.5.6 Duds. Connect the detonator electrically to the test circuit specified and the energy from a capacitor of four (4) microfarads, max, charged to 200 volts, max, shall be discharged to fire the detonator. Observation shall be made for duds.

4.5.7 Non-functioning. Connect the detonator electrically, to the test circuit and apply the energy from a capacitor of four (4) microfarads, max, charged to 25.5 $\pm$ 0.5 volts. Observe if functioning occurs.

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B.

5.1.1.1 Rosette wrap. A bundle of 25 detonators shall be made, lead wires out straight, using the lead wires from one detonator to secure the bundle. Each bundle of detonators shall be wrapped in a sheet of approximately 8 by 8 by .0015 inch thick aluminum foil, complying with specification MIL-A-148, so that all detonators and lead wires in a bundle are completely encased by the foil. Five bundles of detonators, totaling 125 detonators, shall be packaged in a folding carton, set-up box, or comparable container, made of material with a minimum board thickness of 0.3 inch. Each carton, after blunting corners, shall be placed in a water vapor-proof bag manufactured of barrier material in accordance with Type I, Class 3, specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and sealed in accordance with manufacturer's instructions with seal located at edge of bag to provide for minimum of 2 subsequent seals, if required. The seals shall comply with specification MIL-B-117.

5.1.1.2 Pocket assembly (alternate package). The detonators shall be packaged in pocket assemblies, manufactured in accordance with Figure 1 (as applicable), made of .0015 inch aluminum foil complying with specification MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five pockets. The adhesive used for the sealing of the two faces of the aluminum foil shall be suitable for the application. The strength of the seal shall be sufficient to allow the insertion of the dowel (see Figure 1) without separation of the seal. The two faces of any seal shall not be misaligned by more than 1/16 inch when the pocket is formed. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end and similar to that shown on Figure 1. Each pocket may, however, be opened by other suitable methods which will result in providing a satisfactory pocket for insertion of the detonator without injury to the material. One PA526 Electric Detonator shall be inserted, detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to form a closed pouch. Twenty-five pocket assemblies, or a total of 125 detonators, shall be packaged in a folding carton, set-up box, or comparable container, made of a

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material with a minimum board thickness of .03 inch. Each carton, after blunting corners, shall be placed in a water vapor-proof bag, manufactured of barrier material, in accordance with Type I, Class e, specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and sealed in accordance with manufacturer's instruction with seal located at edge of bag to provide for a minimum of 2 subsequent seals, if required. The seals shall comply with specification MIL-B-117.

5.1.2 Level C. Unless otherwise specified, the detonators shall be preserved and packaged as specified in 5.1.1, except that no water vapor-proof protection is required.

5.1.3 Marking. Each inner carton, set-up box, or comparable container, shall be marked: "Patented - US Patent 2,389,086". Marking shall be in accordance with Standard MIL-STD-129 and Department of Transportation (DoT) Code of Federal Regulations 49 CFR 171-179.

5.2 Packing.

5.2.1 Level B (see 6.7). A suitable number of containers, each containing 125 detonator assemblies, shall be packed in outer case in accordance with Tariff #BOE-6000A, Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles, paragraph 173.66 (see Code of Federal Regulations 49 CFR 171-179.

5.2.2 Level C. Unless otherwise specified, the detonators shall be packed as specified in 5.2.1.

5.2.3 Marking. The DoT container marking shall be "Detonators - Handle Carefully". (Apply appropriate explosive label "A"). Marking shall be in accordance with Standard MIL-STD-129 and DoT Code of Federal Regulations 49 CFR 171-179.

5.4 Data cards. Data card information shall be as specified in Specification MIL-A-2550. Classification of the data cards shall be as shown on the applicable Security Requirements Check List.

6. NOTES

6.1 Intended use. The detonator covered by this specification is intended for use for special applications and test programs.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required (see MIL-I-45208, paragraph 3.10).

6.5 Calculation of standard deviation. The following formula, or equivalent, shall be used in calculating the standard deviation:

$$S = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n-1}}$$

Where:

s = standard deviation
x = numerical value of individual observations
x = average value of the sample
n = sample size
= "sum of"

6.6 Check test for possible deterioration (not applicable to new production). The following tests are designed to prevent the loading of PA526 detonators, which have deteriorated, into the major ammunition item. Unless otherwise directed by the Contracting Officer, the tests shall be performed on PA526 detonators from each lot which have been stored under normal conditions for more than two (2) years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the PA526 detonator is loaded does not require that the check tests be performed, the Contractng Officer should insert a provision for the performance of the check test in the contract. In addition, the Contracting Officer should arrange

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to have the contractor reimbursed for any expense incurred in the performance of the check tests for deterioration. Two hundred (200) detonators shall be selected for these tests.

6.6.1 Functioning time (see 3.3.6.1)
Duds (see 3.3.6.1)

Forty (40) detonators shall be subjected to the test specified in 4.5.4. Observation shall bae made for the functioning time. All duds that may occur shall be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\bar{X}$) plus the product of 2.03 times and the standard deviation (s) exceed 8 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.5, or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.6.2 Waterproofness (see 3.3.5)
Output (see 3.3.6.2)
Duds (see 3.3.6.1)

Fifty (50) detonators shall be subjected to the test specified in 4.3.5. Observation shall be made of the depression in the block Duds that may occur shall be recorded. Duds shall be replaced by detonators from the 200 selected. If two detonators fail to adequately damage the block, a retest shall be permitted using the balance of the detonators selected in 6.6. If, during the retest, one more detonator fails to adequately damage the block, the retest shall be stopped and the lot rejected. During the retest, the number of duds that may occur shall be recorded. If three or more detonators fail to adequately damage the block during the original test, the lot shall be rejected without retest.

6.6.3 Duds (see 3.3.6.1). If no retest is required in the output phase, the balance of the detonators selected in 6.6 shall be subjected to the test for duds as specified in 4.5.6. The number of duds that may occur will be recorded. If more than two, but no more than six, duds are found, a retest may be permitted using double the number of detonators selected in 6.6. The lot shall be rejected during retest as soon as the total number of duds observed in 6.6.1, 6.6.2 and 6.6.3 is equal to seven. If, in the original sample of 200, seven or more duds are observed, the lot shall be rejected without retest.

6.7 Level B preservation, packaging and packing. In the event the expected storage time of the detonators is in excess of 180 days, the waterproof barrier bag shall be used as specified in 5.1.1.1 or 5.1.1.2, as applicable. The expected storage time shall be determined by the procuring activity.

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6.8 Data requirements. When this specification is used in acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are involved and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraphs	Data Requirement	Applicable DID
4.4.3.3, 4.4.3.4.1, 4.4.3.4.2, 4.4.3.4.3 4.4.3.5	Quality Inspection Defect Report	DI-R-1721

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